

In a shipyard, a loaded transporter does not drive on the road. It owns the road.

Network-bound capacity, deadlock-free orchestration and fleet-network co-design
(working title)

Road-occupying transport in shipyards: network-bound capacity, deadlock-free orchestration and fleet-network co-design

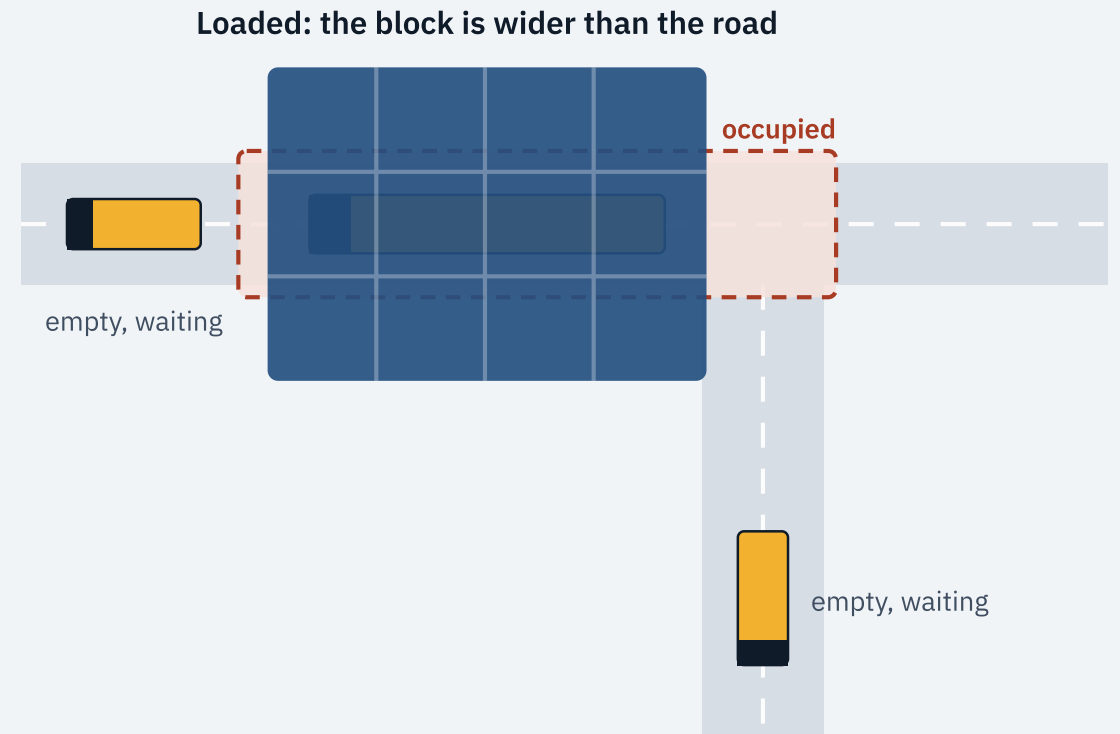
Chunrui Na

Harbin Engineering University · Nanyang Technological University

Claims fixed · three first-order corrections done (10 Oct 2026) · six sub-studies

Free-flow model: vehicles never meet

Occupancy model: loaded vehicles own the road



The whole thesis in one minute

1

The premise

Shipyard block-transport scheduling almost always assumes a free-flow network: travel times from a distance matrix, vehicles that never interact.

2

The question

When loaded vehicles own the road, what sets capacity, and how should fleet, orchestration and network be designed together?

3

Results so far

A rule-independent network bound holds at over 50,000 points in two yards. Okpo's interval [627, 1,720] is floored by whole-route reservation: 27 vehicles at its mean of 339 tasks a day ($< K^* = 50$), 117 at the peak of 678 (16 h).

4

Next

Okpo's new map and deadlock-free rules less conservative than safe release (chapter 4's start); retrofit simulations and the first exchange-rate table; rewriting the proofs of both bounds; the first paper.

Capacity is an interval: the network sets its upper edge, orchestration its lower edge.

The order of this briefing (click to jump)

1

Background and premise

Blocks wider than roads; free flow by default; a gap named by peers p. 4

2

Question, claims, hypotheses

Core question, five claims, seven hypotheses p. 7

3

Model and theorems

Blocking-time model; theorem list and status p. 11

4

Research map

Six sub-studies, each with its own briefing p. 13

5

Results so far

Bound, capacity interval, work zones, dock mouth, retrofits p. 14

6

Papers and venues

Seven chapters, four papers; venues from the school's 2026 list; graduation requirements p. 17

7

Progress and plan

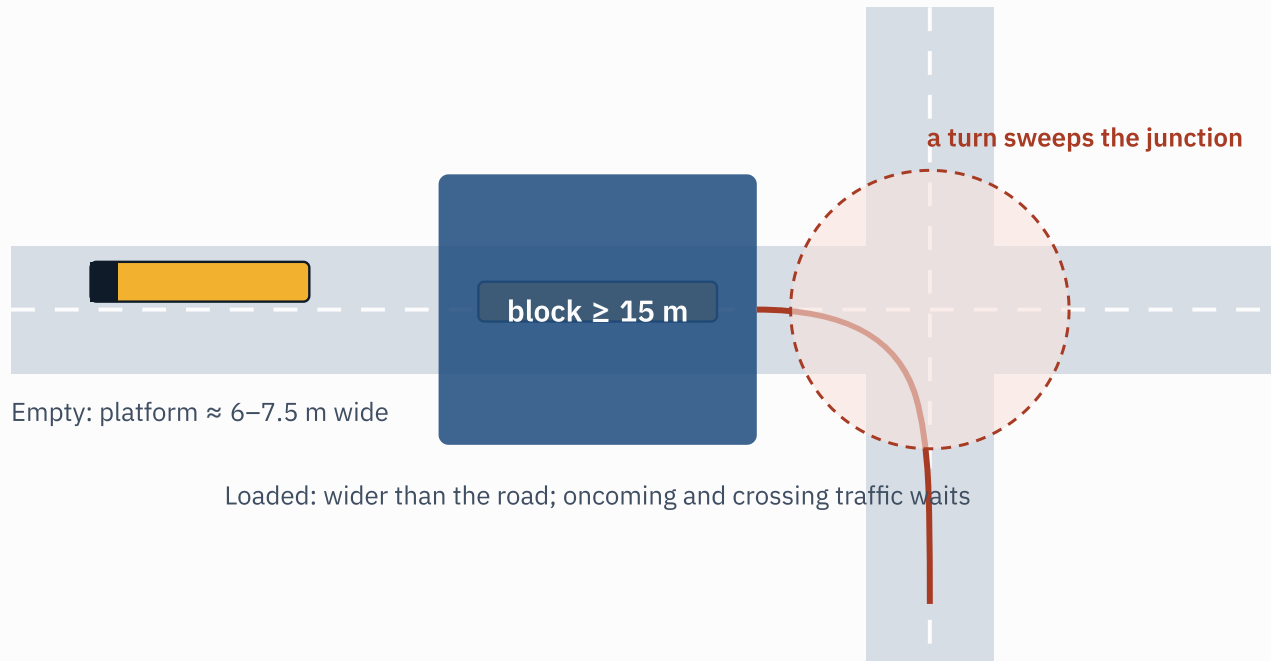
Tasks, eight-week plan, objections, risks and mitigations p. 21

8

Earlier work

How fleet sizing and segmentation serve the thesis p. 26

Blocks are far wider than their carriers: once loaded, a transporter holds the whole segment and junction



- **Empty platforms** are about 6–7.5 m wide; one Hanwha Ocean transporter is 21 m × 7.5 m and drives with signallers ahead and behind
- **Loaded outlines** are often above 15 m, e.g. blocks of about 15 × 15 × 12 m
- **The largest carriers** have platforms of about 30 m × 10 m and occupy the road even when empty

The gap between loaded and empty, and between large and small carriers, is particular to shipyards and weak for AGVs and warehouse robots.

The field's default: a free-flow network in which vehicles never interact

TYPE	WORK	TREATMENT	DISTANCE TO OCCUPANCY
Excluded	Meng et al. 2018	Path overlap in the stockyard only; road interference explicitly excluded	free flow
Free flow	Liu et al. 2022; Zhou et al. 2025; Han et al. 2024	Distance matrices or departure and arrival events only; some recommend fleet sizes on that basis	free flow
Turning delay	Moon et al. 2017	Per-vehicle penalties for turning at junctions, changing each vehicle's shortest path	not occupancy between vehicles
Soft penalty	Roh & Cha 2011; Heo et al. 2013	Interference counted in the objective; penalty when two blocks share a segment at the same time	not a structural capacity constraint; loaded and empty not distinguished; no fleet or network design
Single-yard simulation	Tao et al. 2017	Road capacity and “road interference” in a simulation; utilisation rises after reconfiguring vehicles	mechanism unclear; one yard

No one has treated the exclusive use of segments and junctions by loaded vehicles as a structural capacity constraint, or used it to size fleets or upgrade networks (within our search).

Sources: checked against the originals on 9 Oct 2026. Full texts: Meng 2018, Liu 2022, Zhou 2025 (assumes “traffic congestion not considered”), Roh & Cha 2011 (eqs. 2, 7), Heo 2013, Moon 2017; Han 2024 is a two-page poster, Tao 2017 abstract only. Search scope: two searches of 8 Oct 2026 and the collected literature (shipyard scheduling, multi-agent path finding, railway capacity, queueing networks with blocking).

Peers have named the gap, and the yard data show it

A stated gap: Sun, Liang, Chang (2024), *PLOS ONE*

“how to address the interference limitations of roads in the yard remains a limitation of this study. In actual transportation operations, some roads can only accommodate one transporter at a time.”

Chang Daofang's group, Shanghai Maritime University; they may well follow up, so speed matters.

Yard evidence: Wang, Yin, Khan (2024), *JMSE*

288 delays at a Shanghai yard in 2022: transporters take several lanes and other vehicles must give way; the paper assigns each road-environment event (other vehicles, obstacles, pedestrians) a delay of about 15 min (an assigned value, not a calibration anchor).

Other fields: more vehicles jam, road changes move capacity

- **Container-terminal AGV simulation:** going from 60 to 80 vehicles lowered hourly throughput (by about 2%) Cheng et al. 2005
- **Terminal traffic network:** saturation near 100 vehicles; without keep-clear hatching at junctions, deadlocks from 30 Zhou et al. 2017
- **Underground mine:** single-width roadways with passing bays, trucks managed on railway block-section logic Eustace et al. 2021

These are single-site simulations or operating observations; the shipyard's loaded-to-empty footprint ratio is the most extreme of them.

The core question: when loaded vehicles own the road, what sets capacity?

When loaded vehicles occupy the road exclusively, what determines the capacity of a shipyard block-transport system, and how should fleet, orchestration and network be designed together?

The usual view

- Transport capacity = fleet capacity
- Scheduling means assigning jobs to vehicles
- More vehicles are better, with diminishing returns at worst

The road-occupancy view

- Loaded vehicles hold segments and junctions while passing; loaded and empty occupancy are asymmetric
- Capacity is an interval: the network sets the upper edge, orchestration the lower
- Where the network binds, dispatch becomes road scheduling; at a tight bound, roads can replace vehicles

What sets capacity?

bounds and interval · ch. 3



How to use it fully?

deadlock-free orchestration · ch. 4



How to expand it?

vehicles or roads · ch. 5

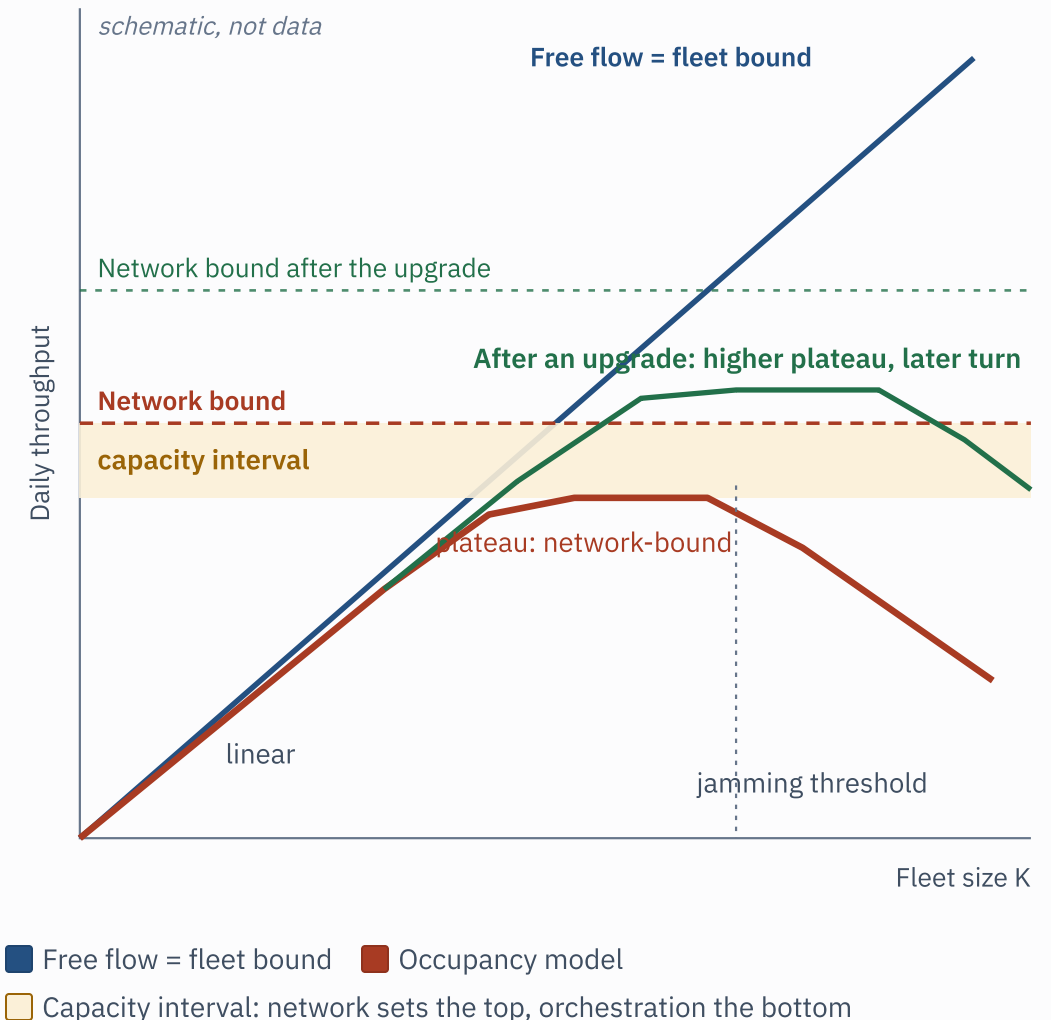


How far does it carry?

cross-yard prediction · ch. 6

Five central claims in one figure

- 1 Two upper bounds and the capacity interval**
 Throughput $\leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{network bound}\}$. The network bound follows from the blocking-time structure and is computable by linear programming. Attainable capacity lies in the interval [best deadlock-free rule, network bound]
- 2 Fleet-throughput trapezoid**
 Throughput first rises linearly with fleet size, then reaches a network-set plateau; when waiting occupies roads it declines past a jamming threshold. Free-flow models see neither of the last two
- 3 Dispatching becomes road scheduling**
 Where the network binds, how much of the bound is attained is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks
- 4 Road-vehicle exchange rate**
 Where the bound is tight, upgrading key elements can replace vehicles, at the dual price converted into vehicles. When the plateau sits far below the bound, change orchestration first. This is where traffic ordering and the exchange rate divide
- 5 Transferable**
 The network bound computed straight from the map, and its binding elements, predict plateau, jamming threshold and upgrade value across yards



Seven hypotheses, each conclusive either way

(1): network-bound, decline, free-flow error

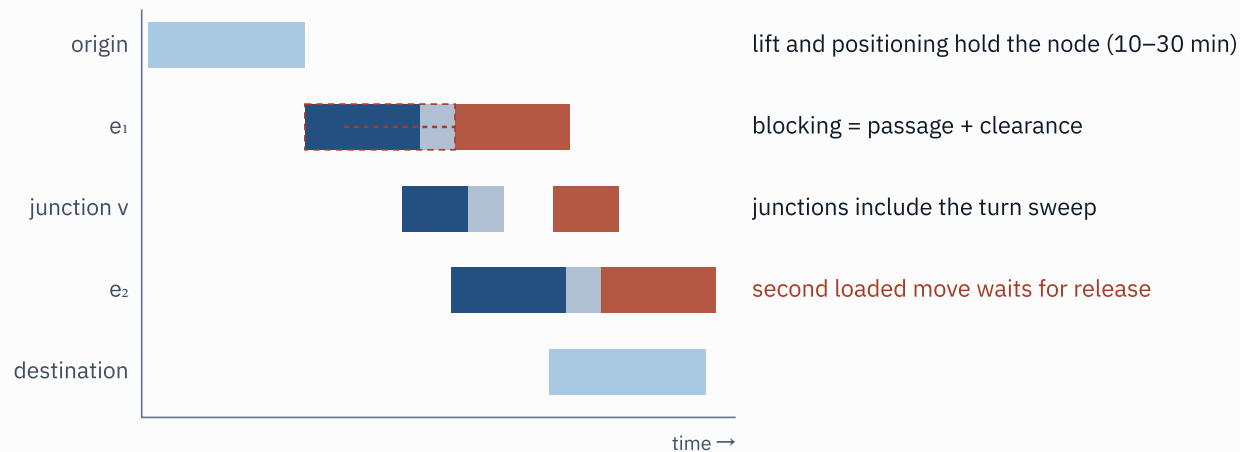
#	HYPOTHESIS	CRITERION	HOLDS / FAILS
1	Network-bound regime: at realistic daily volumes and fleet sizes, the system is network-bound, at least at erection peaks or near the dock mouth	$K^*(20\%)$: marginal gain first and permanently below 20% of free flow; vehicles needed $\geq K^*$ means network-bound; plus “daily volume / network bound”; whole-yard volumes, one conversion for both yards	Network-bound capacity exists / a density criterion for where free flow suffices
2	When the decline appears: with off-road waiting and deadlock-free rules throughput never falls (a plateau); when extra vehicles occupy network resources (waiting on roads, incl. for cranes; finite holding area; physical or no deadlock clearing) it falls beyond K_0 , which moves right with holding capacity S ; good orchestration delays K_0	Read saturated curves under no-teleport rules; scan holding capacity S , clearing mode and where vehicles wait for cranes, one at a time	Conditions under which vehicles hurt, and what a holding area is worth in vehicles / a redundancy index (marginal gain vs free flow)
3	Free-flow error: free flow underestimates completion time and vehicles needed, superlinearly in vehicle density	Service-time gap $\Delta(K) \propto K^\alpha$, $\alpha > 1$; the ratio of vehicles needed read at whole-yard volumes	Systematically wrong at peaks / error under control, with a correction factor

Source: this study's seven hypotheses, fixed on 9 Oct 2026. The network-bound hypothesis is reported in three parts: the capacity interval and its start K^* ; vehicles needed at whole-yard volumes relative to K^* and to free flow; whether the dock mouth caps the yard, which depends on whether vehicles waiting for cranes block the way to other stops.

Seven hypotheses, each conclusive either way (2): upgrade value, safe orchestration, cross-yard prediction, vehicle type

#	HYPOTHESIS	CRITERION	HOLDS / FAILS
4	Road-vehicle exchange rate: where the network binds, one upgrade of a key element is worth at least one vehicle, and dual prices identify the element. Options: segmented blocking, dock-front holding bays, passing bays, one-way loops, widening, a second entrance	Simulated change in vehicles needed vs. the dual-price prediction; agreement is expected only where plateau / bound is near 1	The exchange rate holds / the bottleneck is orchestration; chapter 4 is strengthened
5	Safe orchestration: naive dispatching deadlocks or nearly so at peaks; a provably safe policy loses little throughput	Deadlocks counted under no-teleport rules; safety cost = loss of the best safe rule relative to the network bound	Safe orchestration is cheap / a cost formula for safety and its conditions
6	Cross-yard prediction: the network bound computed from the map, with its binding elements, predicts plateau, K^* and upgrade value across yards; topology explains why the bound binds where it does	Leave-one-out; classes by key element: dock-front single lane, door type ...	A two-second LP screens a new yard / what transfers is the bound tool and the key-element classes
7	Vehicle type and occupancy: large units occupy loaded and empty; small ones yield when empty but run extra empty trips to team up. Case: multi-vehicle moves of giant blocks to floating docks	Compare the best fleet mix across degrees of network constraint	The best fleet mix shifts with network constraint / vehicle type and network can be decided separately

The model: segments and junctions are exclusive resources held for a blocking time



■ loaded move A ■ clearance ■ loaded move B ■ lift, positioning

- **Road graph** $G = (V, E)$: segments e and junctions v are exclusive resources
- **A loaded move** holds each resource on its path for a blocking time, as in railway signalling
- **Empty runs by class**: large units occupy even when empty; small ones can pass on passable links
- **No road widths measured**: passability classes (loaded-exclusive / passable by empties / two-way) are scenario parameters; results are functions of the passable share
- **Resource granularity**: long links can be split at their stops with direction locks; the granularity follows the imagery check of the binding elements

One move holds: $\tau_r = \text{pass}_r + \text{clear}_r, r \in \text{path}$

The theorem list and where each stands

THEOREM	STATEMENT	STATUS
Network bound	The network part of the two bounds: an occupancy-rate LP with width bin-packing rows, crane rows and window-edge terms. Holds for rules meeting the three occupancy conditions; loaded routes fixed	first draft; being rewritten
Route-free bound	Loaded routes are decisions too (multicommodity flow); holds for any routing rule	draft; being rewritten
Trapezoid theorem	On corridors and loops, piecewise throughput in K and the jamming threshold under asymmetric loaded/empty occupancy	to do
Unbounded free-flow error	The free-flow error can be unbounded: a constructive proof	own proof
Non-monotonicity	A single-lane corridor with a holding area of capacity S ; under greedy release throughput strictly falls for $K > K_0(S)$	to do
Safe orchestration	Reserving to the next safe holding point cannot deadlock, with a loss bound relative to the upper bound	in rule comparison
Exchange-rate theorem	The rate comes from dual prices; only upgrades on tight constraints pay. New links use the route-free bound	own proof
Complexity	Feasibility with deadlines and exclusive resources is NP-hard (reduction from blocking job shop)	to do
Candidates	Dock-mouth proposition: with nominal load $p < 1$, blocking-induced starvation destabilises the system; off-road waiting or a segmented dock road restores stability. The other is attainability of the network bound	physical rules support; writing

Source: this study's theorem list and status, fixed on 9 Oct 2026. Highlighted: the theoretical core of the first paper. Proofs and text are written personally.

Research map: six sub-studies

CH. 2–3

Occupancy model and bounds

With loads owning the road, what is the rule-free capacity bound?

two ceilings · network bound · route-free bound

Network bound verified on two yards; system and route-free bounds done

[Open →](#)

CH. 3

Capacity interval and regime

Which interval holds attainable capacity, and when does the network bind?

two ceilings · trapezoid · regime · free-flow error

Capacity interval and work zones done under physical rules; main caliber still to be set

[Open →](#)

CH. 3

The dock mouth at erection peaks

Where do vehicles wait at erection peaks, and can the dock mouth hold up the yard?

network-bound · decline · dock proposition

Holds without teleporting; threshold moves left with fleet size

[Open →](#)

CH. 4

Traffic orchestration

How to order passages without deadlock and push the lower edge up?

traffic ordering · safety cost · safety theorem

Safe release done; less conservative rules being compared

[Open →](#)

CH. 5

Fleet–network co-design

Buy vehicles or change roads? How many vehicles is an upgrade worth?

exchange rate & theorem · upgrade value · vehicle type

Two examples of the ordering / exchange-rate boundary; table from the retrofits

[Open →](#)

CH. 6

Testing across shipyards

Does the bound computed from a map predict across yards?

transfer across yards · cross-yard prediction

Two yards done; maps and base maps ready for 20 yards; group A being traced

[Open →](#)

Ch. 2 defines the object



Ch. 3 where capacity comes from



Ch. 4 using it fully



Ch. 5 spending to expand it



Ch. 6 which yards it reaches

Results so far: both edges of the interval stand, and the dock-mouth effect holds without teleporting

51,767 / 0

The bound holds

None of the 51,767 distinct occupancy points from the gantry-span and physical-rule runs exceeds its network bound. System bound (no sampling noise): Okpo 1,736, Yantai 1,298; the route-free bound holds for any routing rule.

[627, 1,720]

An interval, floored by reservation

Whole-route reservation is the best deadlock-free rule tested; the new safe release is exact and deadlock-free but conservative, below reservation for $K \geq 20$. In both yards the floor is about 36% of the ceiling.

27 / 117

Okpo itself: outside the regime on average, inside at peaks

At Okpo's daily mean of 339 tasks, whole-route reservation needs 27 vehicles, below $K^* = 50$; at the peak of 678 it needs 117 (16-h day). On a 24-h day: 15 and 46.

where to wait

Dock mouth: falls without teleports

Rules that wait for the crane on the dock road, the non-teleporting safe release included, fall as crane load p rises, earlier for larger fleets; reservation, waiting off-road, does not. Splitting the dock road at its stops moves thresholds right.

+76% / +8%

When road changes pay

Erection mix (dock binds): segmenting lifts the network bound 76% and the reservation plateau 8.3%. Main scenario: ceiling +8%, plateau +0.3%. The first rewards road changes, the second orchestration.

both sides

One yard on each side of the test

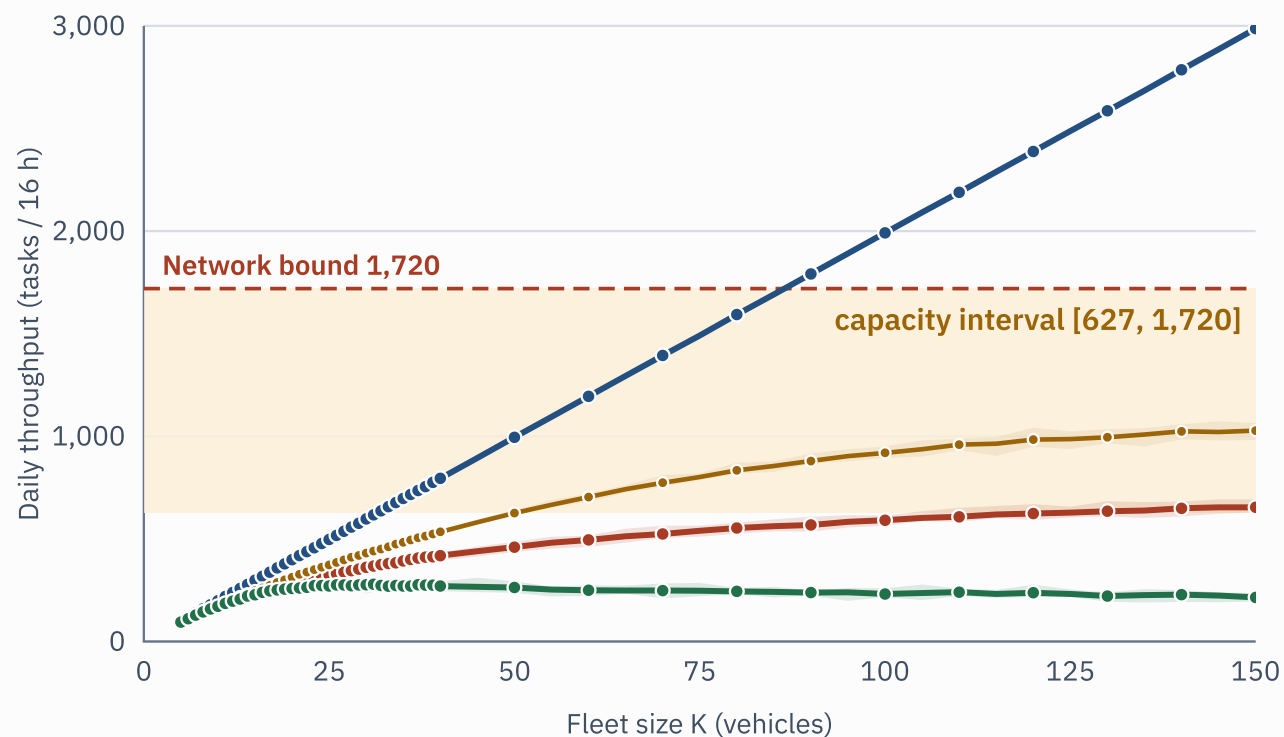
Yantai's own 23–46 tasks a day need 1–2 vehicles; Okpo enters the regime at peaks. Two key-element types, both checked on imagery: a dock-front single lane (Okpo's dock-1 road) and doors (Yantai's presumed paint shop).

Sources: simulations in this study and the imagery check of binding elements. Conditions: the fleet-sizing paper's task mix, Okpo map (before floating docks and branch roads), Yantai map, dock road as one resource (segmenting noted), saturated or finite demand per result; segment request (reference rule, teleport clearing) supports no conclusion on its own. The main caliber of the network-bound test (16 h / 24 h, mean / peak) is set after the comparison with less conservative rules.

The signature figure: free flow keeps rising, occupancy capacity stays inside the interval

Okpo

Yantai



Free flow Whole-route reservation Safe release Segment request (reference)

capacity interval

627-1,720

Capacity interval (a day): lower edge the whole-route reservation plateau (mean of $K = 100-150$), upper edge the network bound

36%

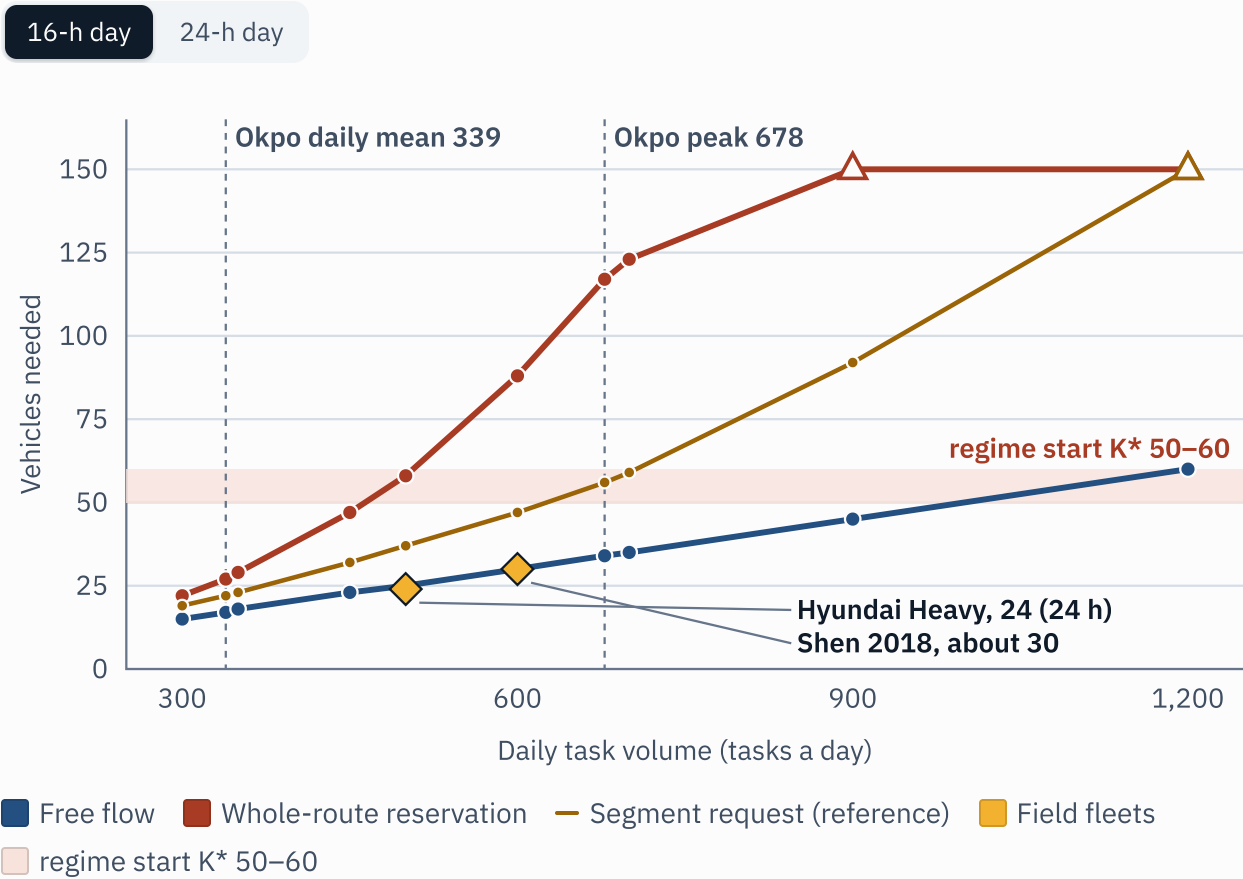
The lower edge reaches only this share of the upper: orchestration decides the rest, which chapter 4 aims to win back

$K \approx 50$

Network-bound from here (marginal ratio first below 20%; range 50-60)

Source: simulation in this study, Okpo map (before floating docks and branch roads) and Yantai map; saturated demand, means of 10 seeds, shading min-max. Throughput = transport tasks per 16-h day. The network bound applies only to the occupancy models; system bound (no sampling noise): Okpo 1,736, Yantai 1,298. Segment request is a reference rule (teleport clearing, 900 s) and supports no conclusion on its own.

Work zones: vehicles needed rise with daily volume, and field fleets sit between free flow and reservation



27

Okpo daily mean of 339: vehicles needed under whole-route reservation, below $K^* = 50$ (16-h day)

117

Okpo peak of 678: vehicles needed under whole-route reservation (16-h day)

30 → 88

600 a day (large Korean yard scale): free flow → whole-route reservation (16-h day)

Source: simulation in this study: Okpo map (before floating docks and branch roads), main scenario, dock road as one resource; vehicles needed = smallest integer K whose 10-seed mean throughput reaches $0.99 \times$ the daily volume; open triangles = not reached at $K = 150$. Okpo's 339 a day is the same-method conversion (294–363 across years, 3% from Hanwha's direct anchor of 350); peak = $\times 2$; Hyundai Heavy about 500 moves, 24 vehicles, 24 h (Park & Seo 2006); Shen et al. 2018 about 600 a day, about 30 vehicles (shifts not stated) (both checked in full text). K^* interval 50–60 from the network-bound test. The main caliber is set after the comparison with less conservative rules.

Thesis plan: seven chapters and four journal papers answering one question

CH.	TITLE	QUESTION	OUTCOME	CLAIMS · HYP.	PAPER
1	Introduction	Where free flow comes from and why it fails in yards	A literature map across four fields	—	—
2	Occupancy model	How to represent wider loads, lifting at nodes, swept junctions	Blocking-time graph; complexity; public-data instance generator	two ceilings	
3	Capacity	Where capacity comes from, when the network binds, when the dock mouth destabilises	Network and route-free bounds; trapezoid, free-flow-error and non-monotonicity theorems; capacity interval and work-zone chart	two ceilings, trapezoid; network-bound, decline, free-flow error	Paper 1
4	Orchestration	How to assign jobs and order passages without deadlock	Exact model, large-scale search, safe-orchestration theorem, rolling online	traffic ordering; safety cost	Paper 2
5	Fleet–network co-design	Buy vehicles or fix roads; large or small vehicles	Exchange-rate theorem and table; control granularity; bilevel algorithm	exchange rate; upgrade value, vehicle type	Paper 3
6	Many yards	Do the laws generalise and predict?	15 public layouts; key-element types; leave-one-out prediction	transfer across yards; cross-yard prediction	Paper 4
7	Conclusion	—	—	—	—

Sources: this study's chapter plan, the assignment of claims and hypotheses, and the planned outputs of chapters 3 and 6. Highlighted: the theoretical core of the first paper.

Where the four papers go: all from the school's 2026 journal list, B-class first, A-class as stretch goals

PAPER	FIRST CHOICE (B)	STRETCH (A) AND ITS CONDITION	ALTERNATIVES (B)	SUPPLEMENTARY (C)
Paper 1 Ch. 2–3 · bound, interval, dock mouth	<i>Transportmetrica A</i> <i>Ocean Engineering</i>	<i>RESS</i> if the dock-mouth proposition becomes a proved result on blocking-induced instability and recovery	<i>IEEE T-ASE</i> <i>Applied Mathematical Modelling</i>	<i>CAIE · JSPD</i>
Paper 2 Ch. 4 · orchestration	<i>IEEE T-ASE</i> <i>Transportmetrica A</i>	<i>IEEE TII · IEEE T-RO</i> rolling online orchestration closed through a digital twin; or provably safe coordination of agents whose footprint grows when loaded	<i>EAAI · ESWA</i> <i>IEEE RA-L</i>	<i>CAIE · AAMAS</i> <i>JMS</i>
Paper 3 Ch. 5 · fleet–network co-design	<i>Ocean Engineering</i> <i>IEEE T-ASE</i>	<i>RESS</i> if written as capacity loss under key-link failure and redundancy design	<i>Transportmetrica A · ESWA</i> <i>ASC · JMP</i>	<i>CAIE · JSPD</i>
Paper 4 Ch. 6 · many yards	<i>Ocean Engineering</i> <i>J. Marine Science and Application</i>	— no natural A-class fit; not forced	<i>EAAI</i> when linked to automatic layout extraction by segmentation	<i>JSPD · IJNAOE</i> <i>JMSE · ISP</i>

Source: the school of shipbuilding's 2026 journal classification (graduation-requirements file, 23 Sep 2026 version, §1 and 7). Transportmetrica A = Transportmetrica A: Transport Science; RESS = Reliability Engineering & System Safety; T-ASE / TII / T-RO / RA-L = IEEE Trans. Automation Science and Engineering / Trans. Industrial Informatics / Trans. Robotics / Robotics and Automation Letters; EAAI = Engineering Applications of Artificial Intelligence; ESWA = Expert Systems with Applications; ASC = Applied Soft Computing; JMP = J. Manufacturing Processes; CAIE = Computers & Industrial Engineering; JSPD = J. Ship Production and Design; AAMAS = Autonomous Agents and Multi-Agent Systems; JMS = J. Manufacturing Systems; IJNAOE = Int. J. Naval Architecture and Ocean Engineering; JMSE = J. Marine Science and Engineering; ISP = International Shipbuilding Progress.

Outputs and graduation requirements: one B-class paper is published, and each thesis paper aims at B-class first

Applying at normal duration

Mid-term evaluation good or better and at least one B-or-above journal paper; then one of: 1 A; 2 B; 1 B + 2 C.

Applying early

Mid-term evaluation excellent and at least one B-or-above journal paper; then one of: 1 A + 1 B; 3 B.

OUTPUT	JOURNAL (CLASS)	STATUS
Segmentation paper	<i>Ocean Engineering</i> (B)	published 368 (2026)
Fleet-sizing paper	<i>Ocean Engineering</i> (B)	target journal
Paper 1 (ch. 2–3)	<i>Transportmetrica A</i> / <i>OE</i> (B) ; <i>RESS</i> (A)	writing, wk 5–8
Paper 2 (ch. 4)	<i>IEEE T-ASE</i> (B) ; <i>TII</i> / <i>T-RO</i> (A)	from the rule comparison
Paper 3 (ch. 5)	<i>OE</i> / <i>IEEE T-ASE</i> (B) ; <i>RESS</i> (A)	from the retrofit runs
Paper 4 (ch. 6)	<i>OE</i> / <i>JMSA</i> (B)	group A being traced

Source: the school of shipbuilding's PhD graduation requirements (2026 journal list update, 23 Sep 2026; output-count rules carried over from the earlier file, superseded by any new school rules) §0, 2, 3; the segmentation paper in its own briefing, the fleet-sizing paper in the block-transporter-choice briefing. JMSA = Journal of Marine Science and Application; other abbreviations on the previous slide.

Algorithms, and what the thesis brings back to three research communities

Algorithms (chapters 4–5)

- **Exact model:** CP-SAT interval variables, one no-overlap constraint per segment and junction; ground truth on small instances
- **Large scale:** integrated assignment-and-sequencing neighbourhood search; priority-based planning for conflicts; compared with “schedule in free flow, then repair conflicts”
- **Safety layer:** the reservation policy of the safe-orchestration theorem as a shell around any heuristic
- **Rolling online:** re-plan under uncertain completion times; release with forecasts
- **Co-design:** outer level picks upgrades and fleet, inner level evaluates by orchestration; dual prices from the exchange-rate theorem rank the outer options
- **Optional frontier:** LLM-evolved orchestration rules (EoH, ReEvo, FunSearch; no shipyard use found)

Framing: “we extend X to Y and find that Z changes under Y”

Multi-agent path finding

A new agent class: footprint multiplies when loaded, long lifts at endpoints, fleet size to be chosen; existing large agents have fixed shapes and no tasks

Railway capacity

Blocking times and compression carried to trackless road networks with asymmetric loaded and empty occupancy

Queueing networks with blocking

A real system that breaks the vehicle–hole symmetry, because empty and loaded vehicles occupy differently

Progress: the three first-order corrections are done; chapter 4's first rule comparison is running

WHEN	ITEM	OUTPUT	STATUS
10-08	Flag experiments on two yards; tighter network bound; erection at crane cadence	Flag curves; dock-mouth mechanism	done
10-09	Full review; claims and hypotheses fixed; imagery check of the two key elements	Three first-order corrections	done
10-09	Literature anchors: Okpo 339 tasks a day; Hyundai Heavy 24 vehicles / 24 h; Tao 2017 full text compared	External comparison; introduction	done
10-10	Safe release rule; segmented dock road; whole-yard volumes (integer K, 16 h / 24 h)	Capacity interval; work zones; dock thresholds by K	done
10-10	System bound; route-free bound; route slack; dual-price table	Bound figures; exchange-rate input	done
from 10-10	Okpo's new map first; less conservative deadlock-free rules: step-wise feasibility search; claims cut at the next safe holding point (safety theorem)	Re-check of the floor, the decline and the safety cost	running
Next	Retrofit simulations (on Okpo's new map); door granularity; three-resource dock network	First exchange-rate table	after rule comparison
Next	Network-bound hypothesis: part 2 numbers, part 3 two-layer wording, main caliber (16 / 24 h, mean / peak)	Final hypothesis wording	after rule comparison
Then	Floating-dock scenario; finite holding area S	Vehicle-type case; decline $K_0(S)$	after retrofits

Source: progress and results of this study's work, as of 10 Oct 2026.

Eight-week plan: the two weeks of corrections are done; the first paper goes out within 3–4 months

WHEN	RESEARCH AND WRITING	COMPUTATION	OUTPUT (PLACE IN THE THESIS)
Weeks 1–2	Claims and hypotheses fixed ✓; imagery checks ✓; group-A tracing	physical rules ✓, bounds ✓, literature anchors ✓; rule comparison (with Okpo's new map) running	Ch. 3 interval, work zones, dock section; bound figures
Weeks 3–4	Rewrite the proofs of both bounds; settle the network-bound hypothesis's three items after the rule comparison; write the dock-mouth proposition (three-resource network)	Retrofits, exchange-rate table; floating docks	First exchange-rate table; ch. 5 prototype; final Okpo map
Weeks 5–8	Write the first paper	External comparison table (normalised by cycle time); network bounds and plateaus for the first group-A yards	Paper 1 draft (Transportmetrica A or OE); first ch. 6 data

The first paper's four results: (1) model and the network bound (system bound, route-free bound); (2) capacity interval and work zones; (3) the dock mouth: where vehicles wait and how the dock road is segmented, holding without teleporting; (4) the rule separating traffic ordering from the exchange rate (two examples under physical rules) and the first exchange-rate table.

Sources: this study's plan for the first paper and the eight weeks; progress (decided 10 Oct 2026: the rule comparison before the retrofits).

The likeliest objections, and the answers

OBJECTION	ANSWER
Yard conflicts are done already Roh & Cha 2011; Heo 2013; Moon 2017; Tao 2017	They treat crossings as soft penalties in path search or turn time for one vehicle; Tao et al. 2017 (full text checked) is a single-yard simulation with road capacity counted in vehicles. None formalises blocking times, compares rules or handles deadlock, gives bounds, or designs fleet and network.
MAPF already has large agents Li et al. 2019	Their shapes are fixed and they carry no pickup-and-delivery tasks; none combines a load-dependent footprint, long lifts and fleet sizing. The thesis uses them as a source to build on.
Yards have too few vehicles to jam	That is what the network-bound hypothesis tests, tested cell by cell under physical rules: at Okpo's own daily mean of 339 the yard is outside the regime under reservation (27 vehicles < 50), at its peak of 678 it is inside (117, 16-h day); at a large yard's 600 a day it holds on a 16-h day too. Either way there is a result.
The dock-mouth drop is a teleport artifact	Tested under physical rules: the non-teleporting, deadlock-free safe release falls with p too, its threshold moving left as fleets grow; reservation, waiting off-road, does not fall; segmenting the dock road moves thresholds right. The result is stated as a conditional reading of where vehicles wait and how the road is segmented, and as a design lever.
No measured data	The theorems need no data; Okpo's daily volume, converted from public data, is 339 (3% from Hanwha's direct anchor of 350); Hyundai Heavy's 24 vehicles (24 h) and Shen et al.'s about 30 both sit between the model's free flow and reservation on a 24-h day.
Exclusivity is too strong	Exclusivity is a parameter: results are functions of the passable share; resource granularity (whole segment, split at stops, doors) is a scenario parameter too, chosen by the imagery check.

Sources: simulations in this study (physical rules and whole-yard volumes); the check of external anchors and the full-text comparison with Tao 2017; Park & Seo 2006; Shen et al. 2018.

Risks and mitigations

RISK	MITIGATION
Working-day length and the mean / peak caliber change the network-bound reading (Okpo's peak holds at 16 h, sits near the start at 24 h)	Report both calipers and state under which each holds; the main caliber is set after the rule comparison. Hyundai Heavy states 24 h and can serve as external evidence.
Safe release is conservative (about 2/3 of its waits come from criterion rejection), so the floor and the decline may change with the rule	A re-check with less conservative deadlock-free rules is running; the floor is always the best rule tested, named, so the interval can only narrow.
Okpo's numbers will shift with the new map (with floating docks and branch roads)	The map is switched before the rule comparison and compared item by item with the earlier map (before floating docks and branch roads); every result names its map, and the paper's figures are redrawn on the final map.
The Chang Daofang group (Sun 2024) follows up	No 2025–2026 follow-up found; the first paper goes out within 3–4 months as planned, with a preprint.
Only two yards so far	All 20 yards on the list now have maps and base maps; group A's six get the network bound first (fast), simulations follow; the imagery check of key elements is a fixed step for every yard.

Sources: this study's risk list (9 Oct 2026); simulations in this study (physical rules and whole-yard volumes); plans for the retrofits and the rule comparison, and the imagery check; candidate yard list (10 Oct 2026).

The results ladder: every chapter's floor is a B-class paper on its own; the ceiling aims at A-class

	FLOOR (DONE ONCE DONE)	TARGET	CEILING
3 Capacity	Two bounds and the trapezoid on one yard two yards in hand	Both bounds and the trapezoid, free-flow-error and non-monotonicity theorems proved; capacity interval and work-zone criterion	Attainability of the network bound and the dock-mouth proposition
4 Orchestration	Exact model and large-scale search beating “free-flow schedule, then repair”	The safe-orchestration policy with small loss; the lower edge pushed to yard per-vehicle productivity	LLM-evolved rules beating hand-written ones and transferring across yards
5 Co-design	Grid evaluation of fleets and upgrades	Exchange-rate theorem and key elements (incl. control granularity)	Bilevel co-design with optimality gaps
6 Many yards	8 yards	15+ yards; key-element types and leave-one-out prediction	A simple formula for jamming thresholds from topology

Earlier work serves as tools; the thesis starts from a new premise

ASSET	USE IN THE THESIS
Fleet-sizing paper	ALNS and CP-SAT code start chapter 4; price data feed chapter 5's vehicle types and costs
Coupled-carriage ideas	Chapter 5, the vehicle-type hypothesis: large carriers against teams under occupancy
Centreline networks and the tracing tool	Production tool for chapter 6's layouts
Remote-sensing segmentation paper (published)	Optional: faster layout extraction for many yards

Open the fleet-sizing briefing →

Open the segmentation briefing →

In a shipyard, a loaded transporter does not drive on the road; it owns the road. Capacity is capped by the network, not the fleet.

Thank you.

Source: this study's plan for earlier assets. The thesis starts from the new premise of road occupancy; earlier work is used as tools and cited.

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